Product Optimization & Revenue Contribution Analysis for Afficionado Coffee Roasters

Executive Summary

Afficionado Coffee Roasters operates in a highly competitive specialty coffee market where understanding product performance is critical for maximizing revenue and improving menu effectiveness. This project analyzes transaction-level sales data to evaluate product popularity, revenue contribution, category performance, and revenue concentration across the product portfolio.

Using Python, SQL, Excel, and Streamlit, a comprehensive analytical framework was developed to identify high-performing products, underperforming menu items, and revenue dependencies. The findings support strategic decision-making related to menu optimization, product promotion, and revenue growth.

The analysis revealed that a relatively small number of products contribute a significant share of total revenue, highlighting opportunities to strengthen high-impact products while reviewing low-performing items for potential redesign or removal.

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1. Introduction

1.1 Business Background

Afficionado Coffee Roasters offers a diverse range of beverages and food products, including coffee, tea, drinking chocolate, bakery items, branded merchandise, and packaged coffee products.

As the product portfolio expands, management faces challenges in understanding:

- Which products customers purchase most frequently
- Which products generate the highest revenue
- How revenue is distributed across categories
- Whether the menu contains low-impact products
- The degree of dependence on specific products or categories

To address these challenges, a data-driven product performance analysis was conducted.

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2. Problem Statement

Although detailed transaction-level sales data is available, Afficionado Coffee Roasters lacks:

- Clear visibility into product popularity versus profitability
- Category-level revenue dependency insights
- Identification of low-impact menu items
- Revenue concentration analysis across the product portfolio
- Decision-support tools for menu optimization

Without these insights, product decisions may rely on assumptions rather than measurable business performance.

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3. Project Objectives

Primary Objectives

- Identify top-selling and least-selling products
- Quantify revenue contribution by product
- Quantify revenue contribution by category
- Measure revenue concentration across the menu
- Compare popularity and profitability

Secondary Objectives

- Support menu simplification initiatives
- Identify high-impact hero products
- Highlight low-performing products for review
- Reduce revenue concentration risk
- Improve overall product portfolio performance

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4. Dataset Overview

Data Source

Internal transaction-level sales dataset.

Key Attributes

- Transaction ID
- Transaction Year
- Transaction Time
- Transaction Quantity
- Store ID
- Store Location
- Product ID
- Unit Price
- Product Category
- Product Type
- Product Detail
- Revenue

Derived Metric

Revenue was calculated using:

Revenue = Transaction Quantity × Unit Price

This metric serves as the foundation for all profitability and contribution analyses.

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5. Data Preparation & Validation

The dataset underwent extensive cleaning and validation before analysis.

Data Quality Checks

- Missing value assessment

```
#Check Missing Values
df.isnull().sum()
[25]
```

- Duplicate transaction detection

```
#Check Duplicate Records
df.duplicated().sum()
[26]
```

```
np.int64(0)
```

- Validate product identifiers and prices

```
#Check Negative or Zero Values
df[df['transaction_qty'] <= 0]
df[df['unit_price'] <= 0]
[36]
```

transaction_id	year	transaction_time	transaction_qty	store_id	store_location	product_id	unit_price	product_category	produc

```
#Remove Invalid Records
df = df[df['transaction_qty'] > 0]
df = df[df['unit_price'] > 0]
[37]
```

- Load Transaction Level Data

```
import pandas as pd
[2]
```

```
df = pd.read_excel("Data/Raw_Afficionado_Coffee_Roasters.xlsx")
[23]
```

Data Transformations

- Revenue calculation
- Product-level aggregation
- Category-level aggregation
- Revenue share calculations
- KPI development

The resulting dataset was transformed into an analysis-ready format suitable for SQL and dashboard reporting.

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6. Methodology

The analytical workflow consisted of five phases:

Phase 1: Revenue Computation

Revenue was calculated for every transaction and aggregated across:

- Product
- Product Type
- Product Category

Phase 2: Product Popularity Analysis

Measured:

- Total units sold
- Product rankings
- Best-selling products
- Least-selling products

Phase 3: Revenue Contribution Analysis

Measured:

- Revenue contribution by product
- Revenue contribution by category
- Product profitability rankings

Phase 4: Category Performance Analysis

Evaluated:

- Category revenue share
- Product-type contribution
- Revenue dependency by category

Phase 5: Revenue Concentration Analysis

Applied Pareto (80/20) analysis to identify:

- Revenue anchors
- Long-tail products
- Revenue concentration risks

7. Key Performance Indicators (KPIs)

Product Revenue Contribution (%)

Measures the percentage of total revenue generated by each product.

Business Value:
Identifies products with the greatest financial impact.

Product Sales Volume

Measures total units sold.

Business Value:
Identifies customer preferences and demand patterns.

Category Revenue Share

Measures revenue generated by each category relative to total revenue.

Business Value:
Evaluates dependence on product categories.

Revenue Concentration Ratio

Measures the percentage of revenue generated by top-performing products.

Business Value:
Assesses business risk associated with revenue dependency.

Product Efficiency Score

Revenue generated per product SKU.

Business Value:
Evaluates product effectiveness and portfolio efficiency.

8. Analysis Results

Product Popularity Findings

Analysis identified products with the highest and lowest sales volumes.

Key observations:

- Several products achieved strong sales frequency.
- Certain products experienced limited customer demand.
- Product popularity varies significantly across categories.

Revenue Contribution Findings

Revenue contribution analysis revealed that:

- A limited number of products generate a substantial portion of total revenue.
- High-volume products are not always high-revenue products.
- Premium products contribute disproportionately despite lower sales frequency.

Category Performance Findings

Category-level analysis indicated:

- Coffee products contribute the largest share of revenue.
- Tea products provide strong supplementary revenue.
- Bakery and specialty products contribute smaller but strategically valuable revenue streams.

This highlights operational dependence on core beverage categories.

Revenue Concentration Findings

Pareto analysis demonstrated that:

- A relatively small percentage of products generate the majority of revenue.
- Long-tail products contribute minimal revenue.
- Excessive menu complexity may not provide proportional financial value.

This suggests opportunities for menu optimization.

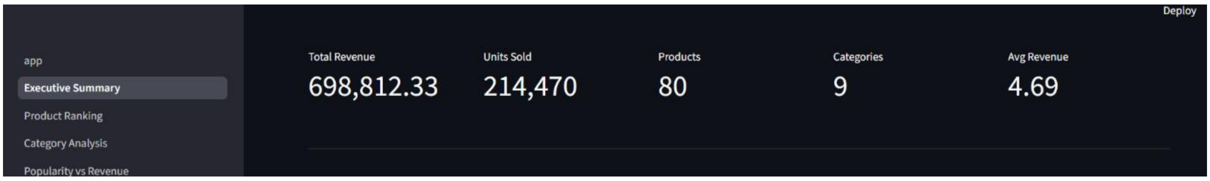
9. Dashboard Development

A multi-page Streamlit dashboard was developed to support business users.

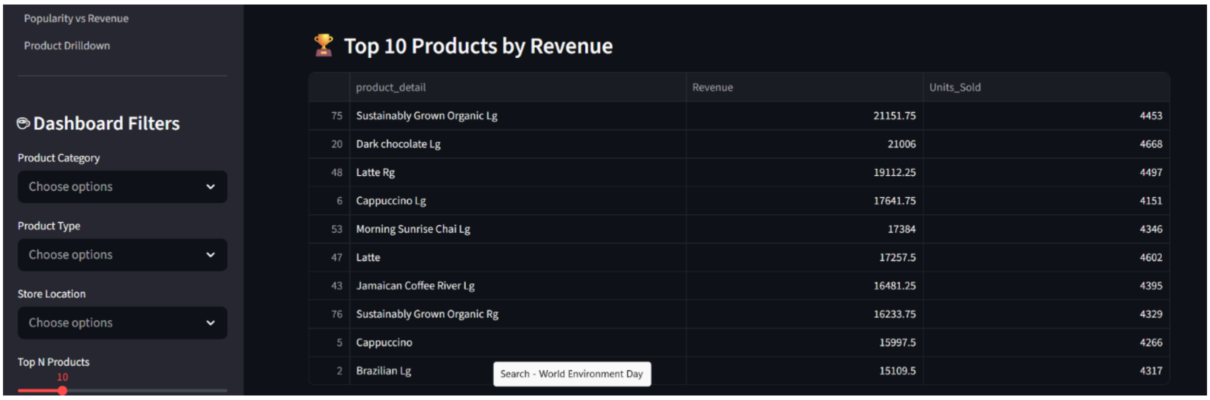
Modules

Executive Summary

- KPI overview

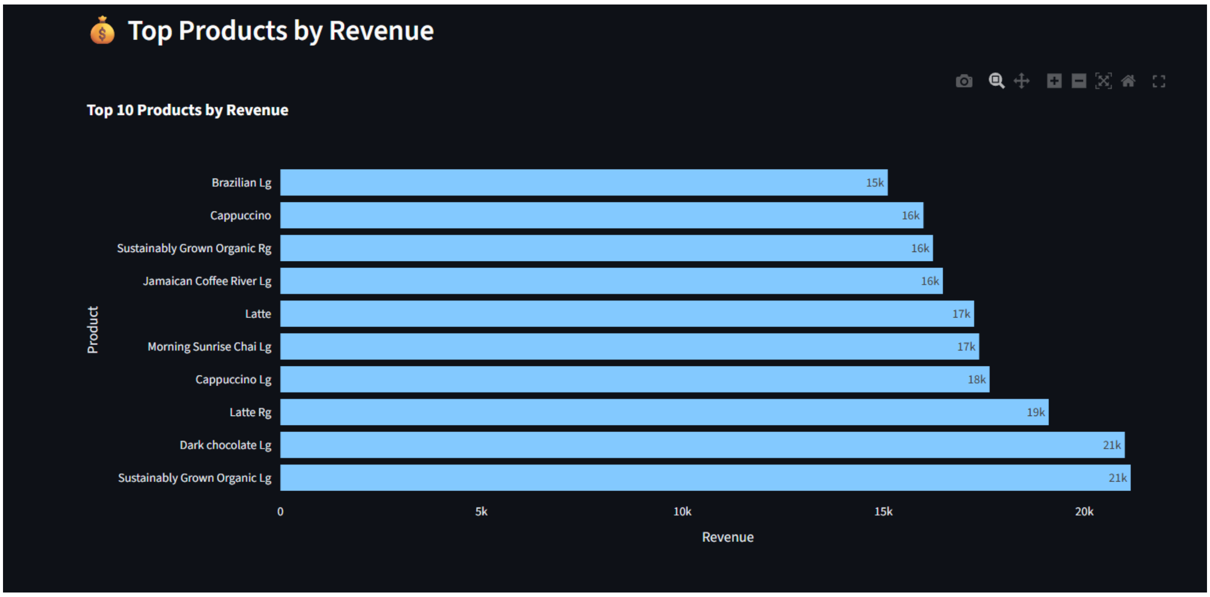


- Revenue performance snapshot

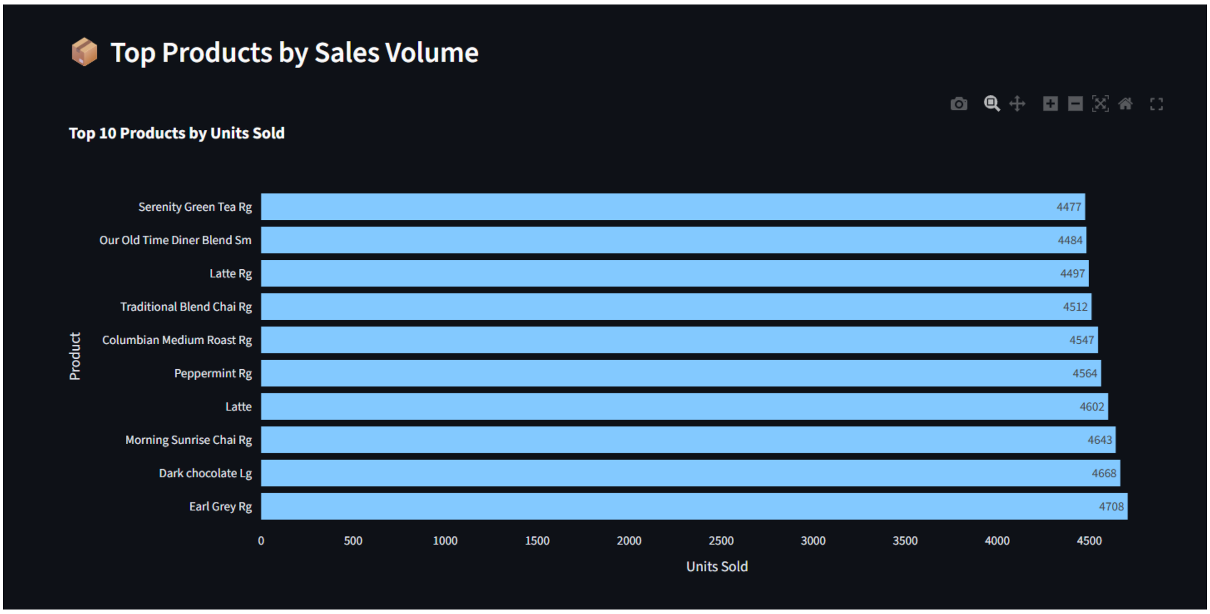


Product Ranking

- Revenue ranking

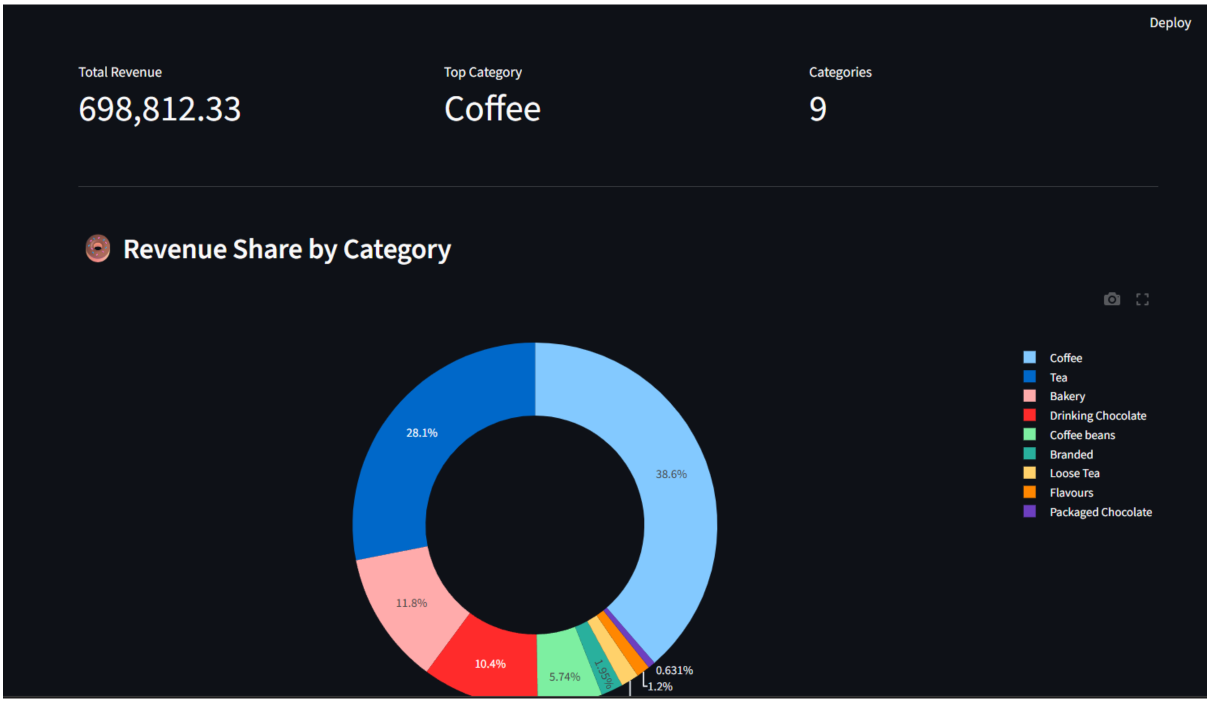


- Sales volume ranking

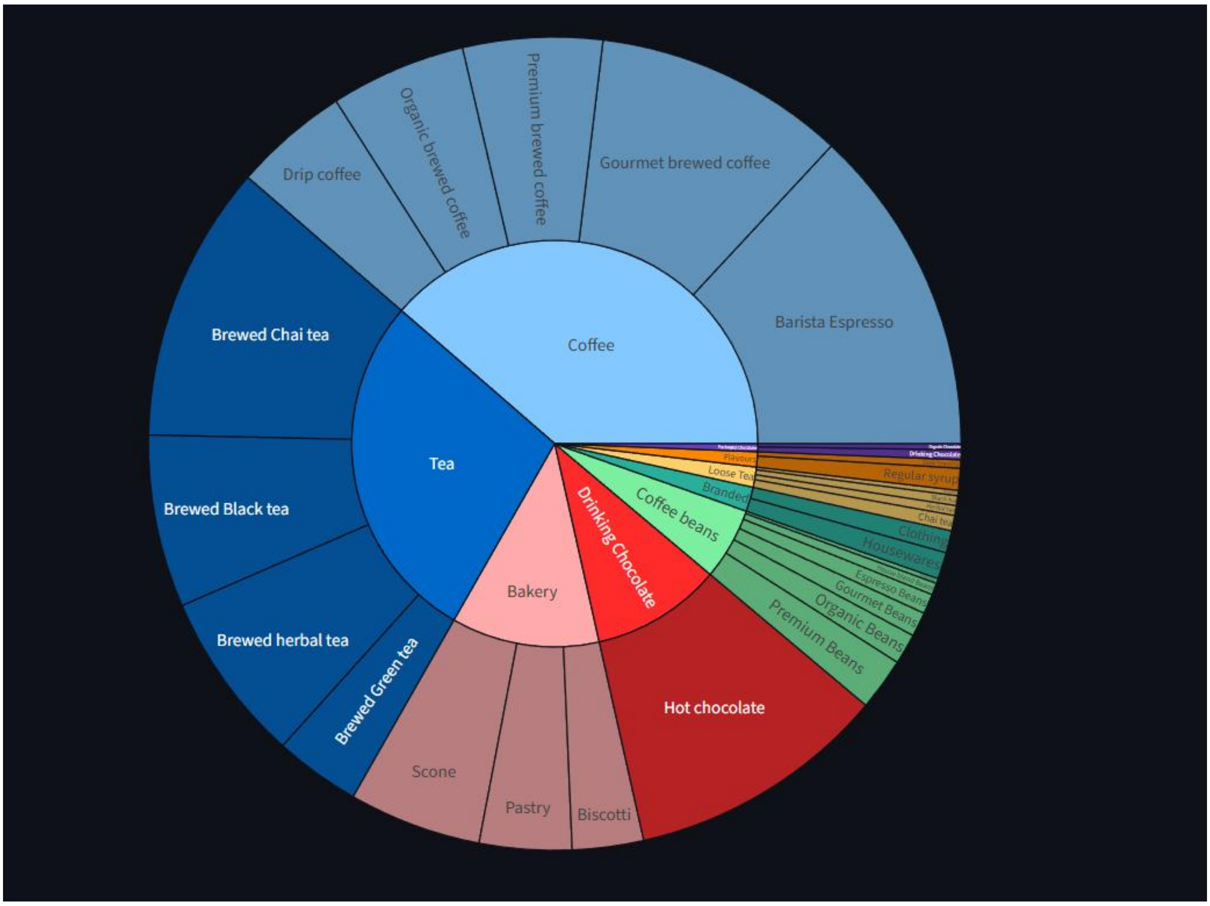


Category Analysis

- Revenue distribution



- Category dependency analysis



Popularity vs Revenue

- Product performance comparison

📋 Product Comparison Table

	product_detail	Units_Sold	Revenue	Avg_Price
75	Sustainably Grown Organic Lg	4453	21151.75	4.75
20	Dark chocolate Lg	4668	21006	4.5
48	Latte Rg	4497	19112.25	4.25
6	Cappuccino Lg	4151	17641.75	4.25
53	Morning Sunrise Chai Lg	4346	17384	4
47	Latte	4602	17257.5	3.75
43	Jamaican Coffee River Lg	4395	16481.25	3.75
76	Sustainably Grown Organic Rg	4329	16233.75	3.75
5	Cappuccino	4266	15997.5	3.75
2	Brazilian Lg	4317	15109.5	3.5

- Hero product identification

🏆 Top Revenue Products

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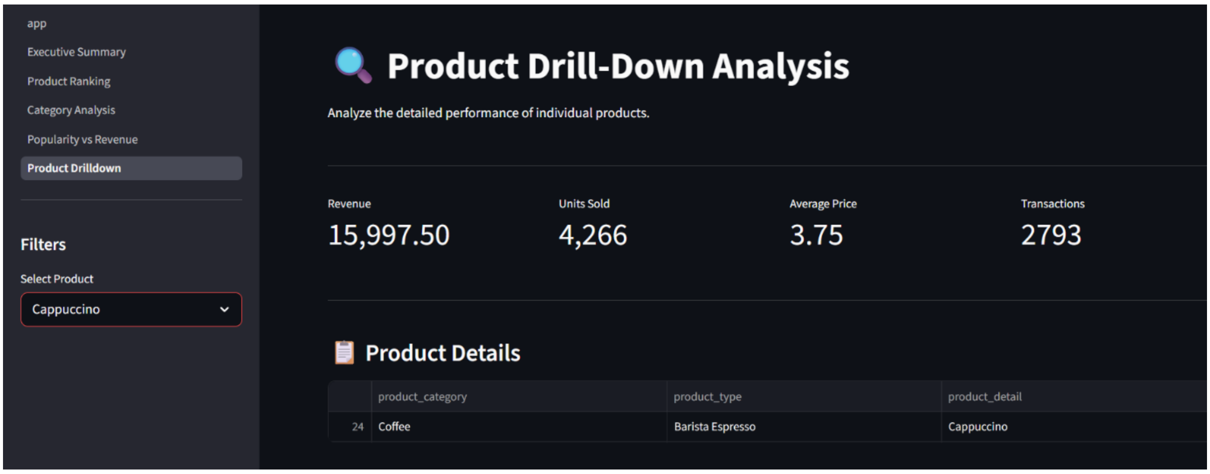
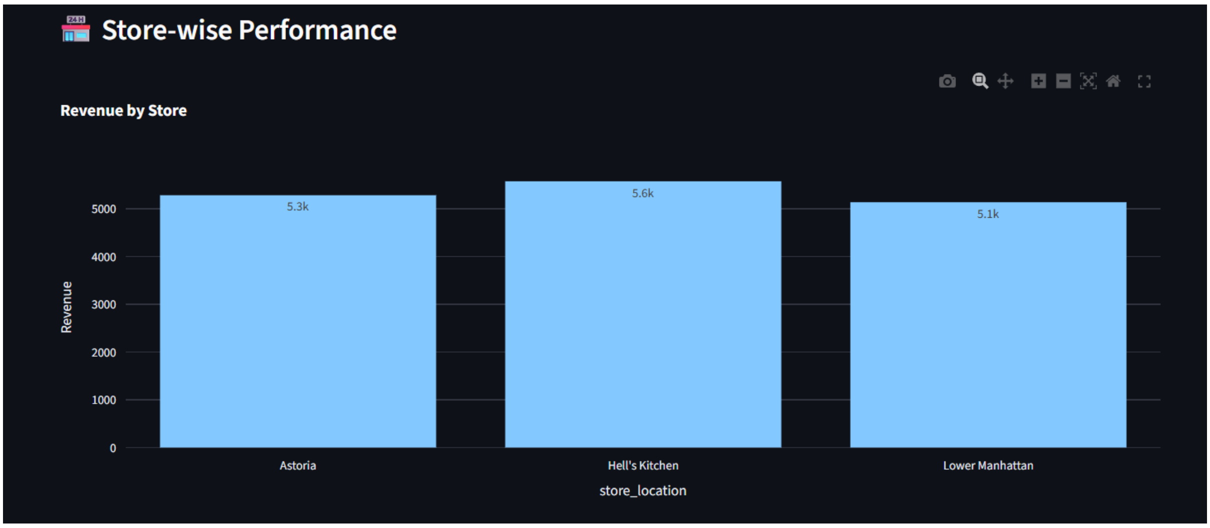
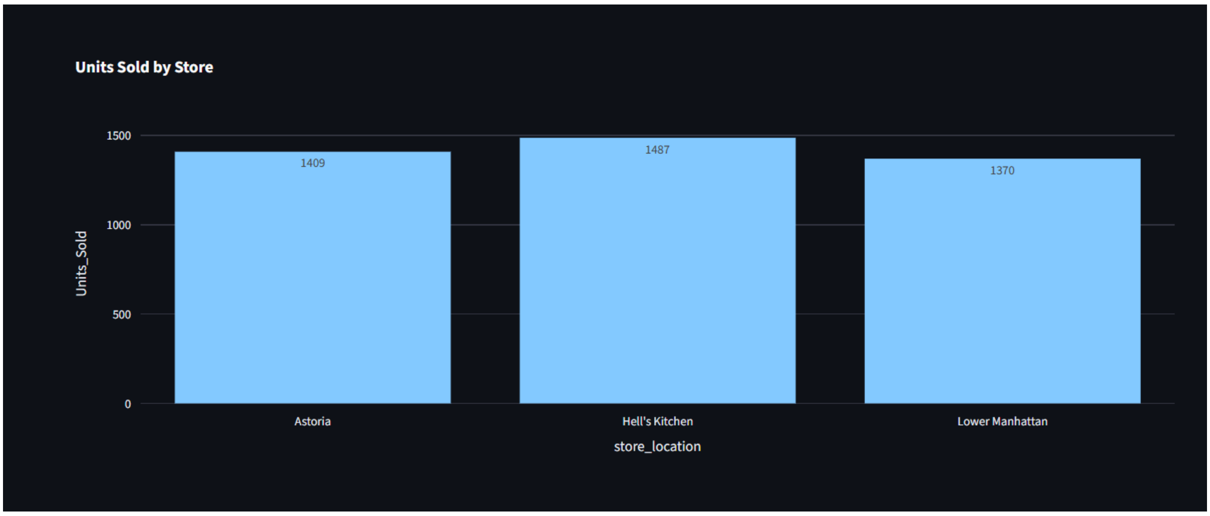
Product Drill-Down

- Product-level performance



- Store-level analysis





📊 Store Performance Table

	store_location	Revenue	Units_Sold
0	Astoria	5283.75	
1	Hell's Kitchen	5576.25	
2	Lower Manhattan	5137.5	

📊 Store Performance Table

	store_location	Revenue	Units_Sold
0	Astoria	5283.75	1409
1	Hell's Kitchen	5576.25	1487
2	Lower Manhattan	5137.5	1370

User Features

- Category filters
- Product-type filters
- Store location filters
- Top-N product selection

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10. Business Insights

Hero Products

Products exhibiting both high revenue and high sales volume should receive increased promotional support.

Premium Products

Products generating high revenue with lower sales frequency may benefit from targeted marketing.

Underperforming Products

Low-revenue, low-volume products should be reviewed for redesign, repositioning, or removal.

Category Dependency

Heavy dependence on a small number of categories may increase revenue risk and reduce diversification.

11. Recommendations

1. Increase marketing investment in top revenue-generating products.
2. Introduce bundle offers around high-performing products.
3. Review long-tail products with consistently low contribution.
4. Monitor category dependency to reduce revenue concentration risk.
5. Use dashboard insights as part of monthly product performance reviews.
6. Continuously track product performance using automated reporting.

12. Conclusion

This project successfully transformed transaction-level sales data into actionable business intelligence. Through revenue contribution analysis, product ranking, category evaluation, and revenue concentration assessment, Afficionado Coffee Roasters gained visibility into the drivers of sales performance and profitability.

The resulting dashboard provides management with a scalable decision-support system capable of improving menu effectiveness, reducing operational complexity, and supporting sustainable revenue growth.